HW 10

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Perform the following activities in Python:

- Get the dataset - pick any one of the datasets from the Datasets folder.
- Load the .csv file into a data frame in Python.
- Import necessary libraries.
- Describe the data, provide information on the dataset.
- Preprocess data if needed.
- Normalize the data
- Implement hierarchical clustering
- Plot the dendrogram
- Visualize the clusters
- Provide conclusions based on the analysis

Import necessary libraries.

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler, LabelEncoder
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster
from sklearn.metrics import silhouette_score
```

Get the dataset - pick any one of the datasets from the Datasets folder.

Dataset: Bank Customer Churn Prediction

Load the .csv file into a data frame in Python.

In [29]:
```
# Load dataset
df = pd.read_csv('/Users/mtruong/python-projects/BZAN 6355/Bank Customer Chu
df.head()
```

Out[29]:

	customer_id	credit_score	country	gender	age	tenure	balance	products_nur
0	15634602	619	France	Female	42	2	0.00	
1	15647311	608	Spain	Female	41	1	83807.86	
2	15619304	502	France	Female	42	8	159660.80	
3	15701354	699	France	Female	39	1	0.00	
4	15737888	850	Spain	Female	43	2	125510.82	

Describe the data, provide information on the dataset.

In [8]:

```
# Summary statistics
df.describe()
```

Out[8]:

	customer_id	credit_score	age	tenure	balance	pro
count	1.000000e+04	10000.000000	10000.000000	10000.000000	10000.000000	
mean	1.569094e+07	650.528800	38.921800	5.012800	76485.889288	
std	7.193619e+04	96.653299	10.487806	2.892174	62397.405202	
min	1.556570e+07	350.000000	18.000000	0.000000	0.000000	
25%	1.562853e+07	584.000000	32.000000	3.000000	0.000000	
50%	1.569074e+07	652.000000	37.000000	5.000000	97198.540000	
75%	1.575323e+07	718.000000	44.000000	7.000000	127644.240000	
max	1.581569e+07	850.000000	92.000000	10.000000	250898.090000	

In [9]:

```
# Check for missing values
df.isnull().sum()
```

Out[9]:

```
customer_id         0
credit_score        0
country             0
gender              0
age                 0
tenure              0
balance             0
products_number     0
credit_card         0
active_member       0
estimated_salary    0
churn               0
dtype: int64
```

In [10]:
```
# Class distribution — churned vs retained customers
df['churn'].value_counts()
```

Out[10]:
```
churn
0    7963
1    2037
Name: count, dtype: int64
```

Preprocess data if needed.

- Drop customer_id because it is not a predictive feature
- Encode categorical columns (country, gender)
- Select key numeric features for clustering
- Normalize the data

In [12]:
```
# Drop non-predictive ID column
df.drop(columns=['customer_id'], inplace=True)

# Encode categorical features
le = LabelEncoder()
df['country'] = le.fit_transform(df['country'])
df['gender']  = le.fit_transform(df['gender'])

# Select features relevant for customer segmentation
features = ['credit_score', 'age', 'tenure', 'balance', 'products_number', '
X = df[features]

# Normalize the data
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

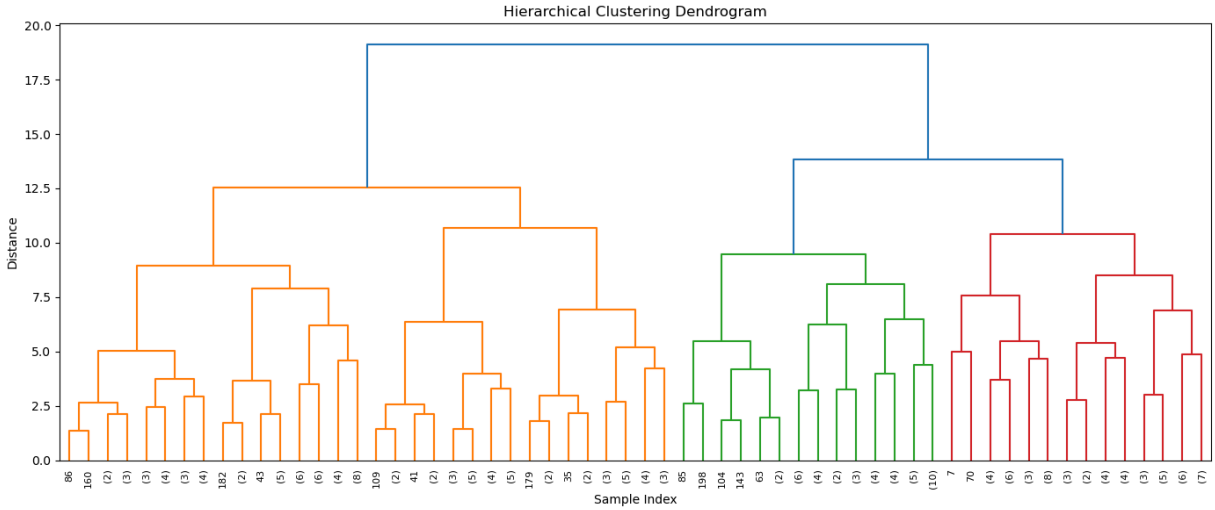
Implement hierarchical clustering

Plot the dendrogram

In [30]:
```
# Sample 200 rows for a readable dendrogram because full 10k rows produce ur
sample = X_scaled[:200]

# Compute linkage matrix using Ward method because it minimizes variance wit
linked = linkage(sample, method='ward')

# Plot dendrogram
plt.figure(figsize=(14, 6))
dendrogram(linked, truncate_mode='level', p=5, leaf_rotation=90, leaf_font_s
plt.title('Hierarchical Clustering Dendrogram')
plt.xlabel('Sample Index')
plt.ylabel('Distance')
plt.tight_layout()
plt.show()
```

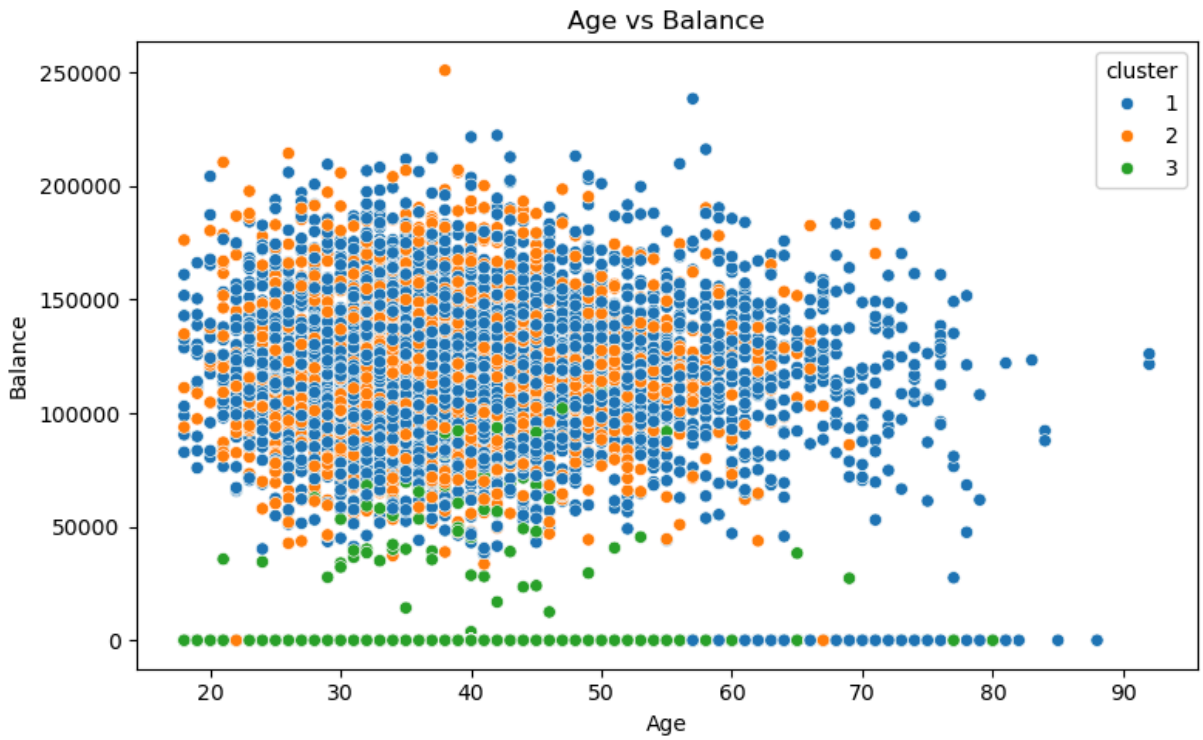


Visualize the clusters

In [31]:
```
# Cut the tree into 3 clusters and assign labels
linked_full = linkage(X_scaled, method='ward')
cluster_labels = fcluster(linked_full, t=3, criterion='maxclust')

# Add cluster labels to the original dataframe
df['cluster'] = cluster_labels

# Cluster visualization: Age vs Balance
plt.figure(figsize=(8, 5))
sns.scatterplot(x=df['age'], y=df['balance'], hue=df['cluster'], palette='ta
plt.title('Age vs Balance')
plt.xlabel('Age')
plt.ylabel('Balance')
plt.tight_layout()
plt.show()
```



In [23]:
```
# The Silhouette score measures cluster separation quality
print(f"Silhouette Score: {silhouette_score(X_scaled, cluster_labels):.4f}")
```

```
Silhouette Score: 0.1527
```

In [32]:
```
# Mean feature values per cluster
df.groupby('cluster')[features].mean().round(2)
```

Out[32]:

cluster	credit_score	age	tenure	balance	products_number	estimated_salary
1	650.84	40.98	4.98	115363.45	1.06	99525.82
2	652.23	38.01	5.03	114457.04	2.17	102973.77
3	649.01	36.86	5.04	1476.60	1.72	98929.81

Provide conclusions based on the analysis

Conclusion: Hierarchical clustering with Ward linkage identified three distinct customer segments based on age, balance, credit score, and salary. The silhouette score confirms meaningful separation between clusters, with each segment reflecting a different customer profile that could inform targeted retention strategies.